

Octo · arXiv 2405.12213 · RSS 2024

Archaeologist

11+3

ACT||DP → Octo → OpenVLA/ π_0
→ OpenPI/ $\pi_{0.5}$ /RTC → BEHAVIOR'25-'26

Oct 14 paper club · VLA Architectures · Oct 16 BEHAVIOR nearby

Spine: open OXE GRP + diffusion chunks · underspecified horizons · NOT 2026 winner weights

Octo Model Team · Berkeley / Stanford / CMU / DeepMind · HF rail-berkeley/octo-*

Genealogy · CLEAR club line



Pivot: open ViT + diffusion-chunk GRP on 800k OXE with block-wise adapters

Octo = infrastructure history + eval baseline · NOT 2026 winner weights

Priors: OXE/RT-X · DP · ACT · ViT · T5 · DDPM · HER goals · RT-1/RoboCat · Bridge

Subsequent CLEAR: OpenVLA · π_0 (2410.24164) · OpenPI · $\pi_{0.5}$ /RTC · BEHAVIOR'25-'26

Club line

**ACT || DP → Octo (open OXE GRP)
→ π_0 / OpenVLA → OpenPI / BEHAVIOR**

π_0 (CLEAR) beats Octo-93M on hard dexterous tasks with VLM+FM.
Octo = infrastructure history + eval baseline — not 2026 winner weights.

Prefer $\pi_{0.5}$ / N1.7 for BEHAVIOR · keep Octo for open FT ancestor + mixture lessons

Steelman · what the paper actually delivers

Open GRP stack

Weights + train + loaders
Small/Base ckpts · HF
RT-2-X / RoboCat did not match

Compositional I/O

Block-wise attention
Mask missing cams/lang
Add lightweight adapters at FT

Diffusion chunks

ViT-first + DDPM head
Imports ACT/DP control culture
Table II: diffusion $\gg$ MSE/discrete

Operational FT recipe

~100 demos · 50k steps
~5 h on A5000 24 GB
Shared HPs across 6 domains

Integrative + open systems novelty — not a single new module

Cite openness + finetune API as the primary artifact,
not “beats RT-2-X” alone. Component pieces (transformer, OXE, diffusion, chunks) each existed.

Octo stack · ViT + frozen T5 + DDPM chunk head

Single backbone forward · light head denoises · receding-horizon execute (TE not helpful)

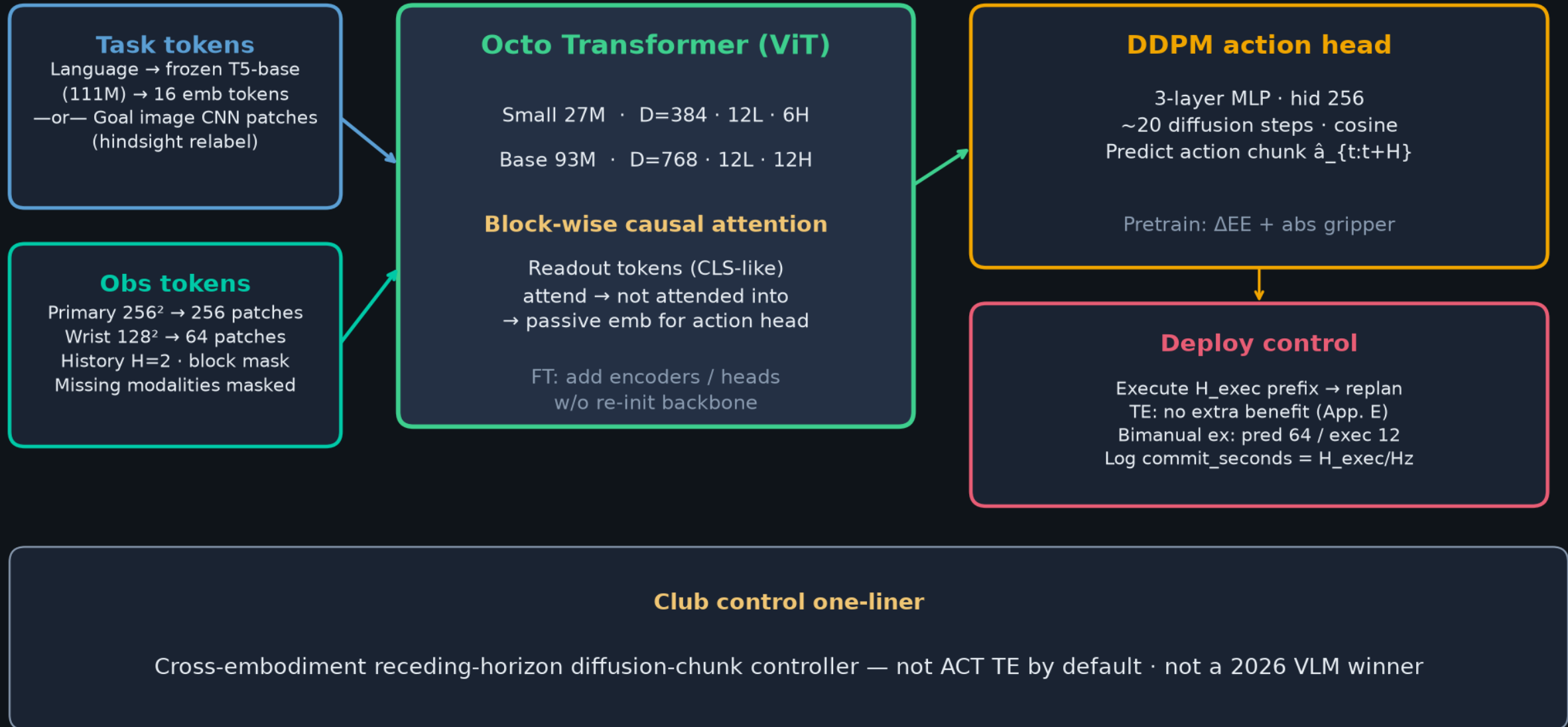


Fig. 2 · Model architecture — tokens → transformer → readout → action head

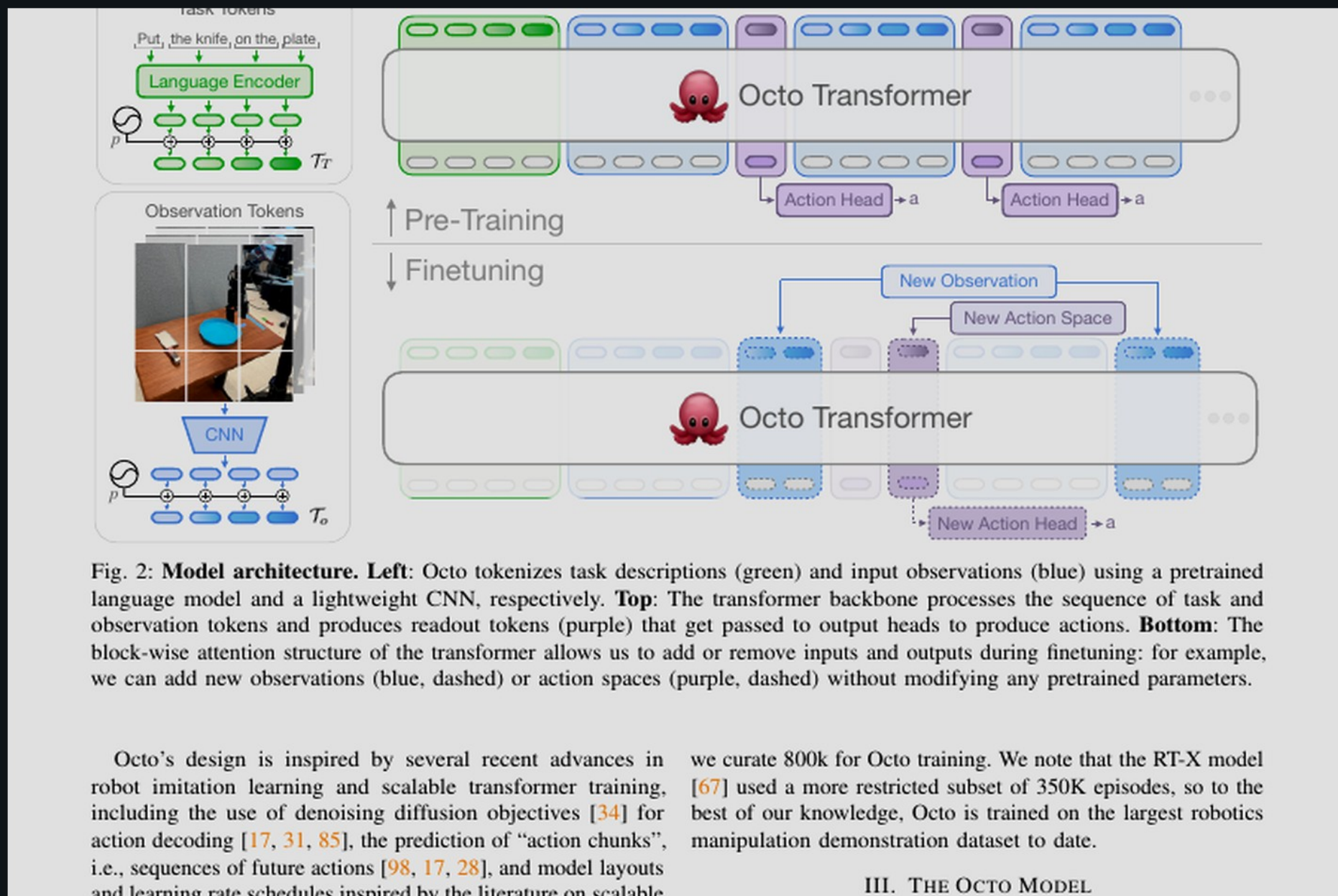


Fig. 2: **Model architecture.** **Left:** Octo tokenizes task descriptions (green) and input observations (blue) using a pretrained language model and a lightweight CNN, respectively. **Top:** The transformer backbone processes the sequence of task and observation tokens and produces readout tokens (purple) that get passed to output heads to produce actions. **Bottom:** The block-wise attention structure of the transformer allows us to add or remove inputs and outputs during finetuning: for example, we can add new observations (blue, dashed) or action spaces (purple, dashed) without modifying any pretrained parameters.

Octo's design is inspired by several recent advances in robot imitation learning and scalable transformer training, including the use of denoising diffusion objectives [34] for action decoding [17, 31, 85], the prediction of "action chunks", i.e., sequences of future actions [98, 17, 28], and model layouts and learning rate schedules inspired by the literature on scalable

we curate 800k for Octo training. We note that the RT-X model [67] used a more restricted subset of 350K episodes, so to the best of our knowledge, Octo is trained on the largest robotics manipulation demonstration dataset to date.

III. THE OCTO MODEL

Underspecified horizons · motion FLAG

Main text: “several” future actions · only concrete: bimanual pred 64 / exec 12

System	Chunk object	Generative	Ensemble?	Typical commit
ACT	k×14 joints	CVAE	Yes TE	ensembled @ 50 Hz
Diffusion Policy	Tp actions	DDPM/DDIM	warm / Ta	Ta≈8 @ ~10 Hz
Octo (this)	action chunk	DDPM readout	No (paper)	receding prefix
π_0 / $\pi_{0.5}$	H actions	Flow matching	No	open-loop K @ 20-50
BEHAVIOR'25	H≈30-32	FM + corr.	soft inpaint	~30 Hz receding

Empiricist: always log (H_pred, H_exec, Hz, diff_steps) · commit_seconds = H_exec/Hz · TE not dead globally

Fig. 3 · Training mix — 25 OXE datasets

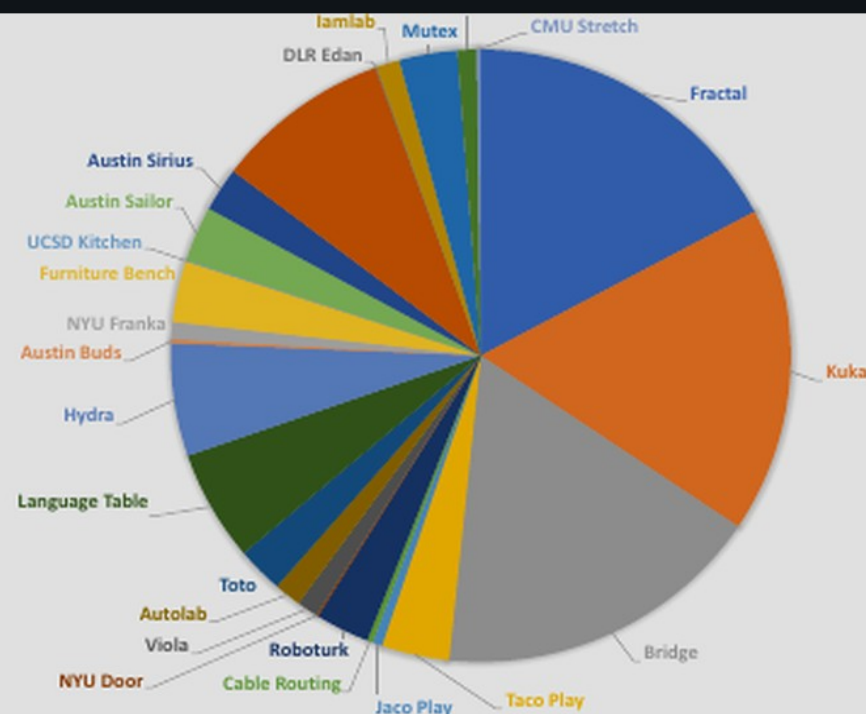


Fig. 3: **Training dataset composition.** We curate a subset of 25 datasets from the Open X-Embodiment dataset that have image observations, end-effector actions, and show diverse behaviors. The pie chart visualizes the fractions that each dataset contributes to every training batch on average. The dataset weights are determined by the number of samples in each dataset with small modifications to balance dataset size and diversity (see [Section III-B](#) for details).

action configurations during pretraining, being able to adapt Octo's inputs and outputs during finetuning makes it a versatile tool for the robotics community. Prior model designs that use

$x^k \sim \mathcal{N}(0, I)$ and apply k steps of denoising with a learned denoising network $\epsilon_\theta(x^k, e, k)$ that is conditioned on the output x^k of the previous denoising step, the step index k , and the output embedding e of the transformer action readout:

$$x^{k-1} = \alpha(x^k - \gamma\epsilon_\theta(x^k, e, k) + \mathcal{N}(0, \sigma^2 I)). \quad (1)$$

The hyperparameters α , γ , and σ correspond to the noise schedule: we use the standard cosine schedule from [66]. We train the diffusion head using the standard DDPM objective first proposed in [34], where we add Gaussian noise to the dataset actions and train the denoising network $\epsilon_\theta(x^k, e, k)$ to reconstruct the original action. For a detailed explanation of diffusion policy training, see Chi et al. [17]. We list all hyperparameters in Appendix D.

We use the same diffusion training objective during finetuning and update the full model, a recipe which outperformed those that freeze subsets of the pretrained parameters. In all finetuning experiments, we employ the same recipe: given a small target domain dataset with around 100 trajectories, we finetune for 50k steps using a cosine decay learning rate decay with linear warmup.

D. Training Details

We trained two variants of our model: Octo-Small with a transformer backbone that mirrors the size of a ViT-S, and Octo-Base with a transformer backbone that mirrors the size of a ViT-B [22].

We use the AdamW optimizer [51] with an inverse square root decay learning rate schedule [97], with weight decay of

- **Finetuning scripts** for Octo models, in JAX.
- **Model pretraining pipeline** for Octo pretraining on the Open X-Embodiment dataset, in JAX.
- **Standalone data loaders** for Open X-Embodiment data, compatible with JAX and PyTorch.

We provide a simple example for loading and running a pretrained Octo model in Appendix B.

IV. EXPERIMENTS

Our experiments provide an empirical analysis of Octo, evaluating its ability to serve as a general robotic foundation model across several axes:

- 1) Can Octo control multiple robot embodiments and solve language and goal tasks out of the box?
- 2) Do Octo weights serve as a good initialization for data-efficient finetuning to new tasks and robots, and does it improve over training from scratch and commonly used pretrained representations?
- 3) Which design decisions in Octo matter most for building generalist robot policies?

Evaluation setups: We evaluate Octo’s capabilities across a representative spectrum of 9 robot learning setups at 4 institutions (see Fig. 4). We test Octo’s ability to control different robots out-of-the-box (“zero-shot”) for language and goal image tasks using robot setups that match the pretraining data, where all robots are controlled with delta end-effector control actions and the observation spaces are RGB images. We also evaluate Octo for data-efficient finetuning to new environments and

Octo on 9 real robot setups across 4 institutions. Our evaluations capture diverse object long task horizons (e.g., “Stanford Coffee”) and precise manipulation (e.g., “Berkeley capabilities to control robots in environments from the pretraining data out-of-the-box and environments with small target domain datasets. We also test finetuning with new Berkeley Peg Insertion”), action spaces (joint position control in “Berkeley Pick-Up” and bodiments (e.g., “Berkeley Bimanual” and “Berkeley Coke”).

ise manipulation (e.g., each evaluation setup,

’s ability to control best openly available using the released pretrained on the Open to control multiple point of comparison. ities of Octo to RT- mage model finetuned produce robot actions. re trained on a more are to 800k episodes performance as a policy ng to two common main demonstrations isual representations. osed other pretraining [26], to our knowledge policy that has been new observation and representations such

Majumdar et al. [57]. A ViT-B visual encoder is initialized to the VC-1 weights [57], a state-of-the-art visual representation pretrained on 4,000 hours of ego-centric videos and ImageNet, and combined with an MLP action decoder. The full model is trained to predict expert actions using an MSE loss (“VC-1”).

A. Octo Controls Multiple Robots Out-of-the-Box

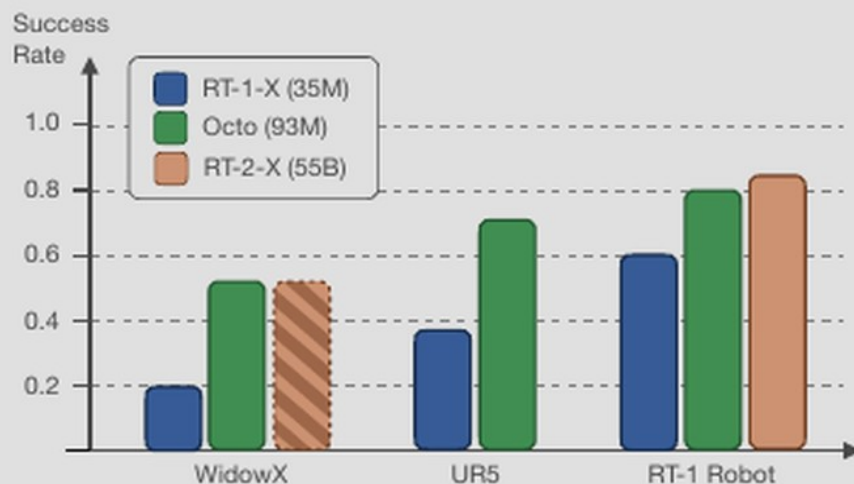


Fig. 5: **Zero-Shot Evaluation.** Out-of-the-box, Octo can control multiple robots in environments from the pretraining data.

rates are averaged over 10 trials on one language-conditioned task per robot.

by comparing to Octo models trained on a smaller mix of 11 datasets used in training the RT-X models [67] and a baseline trained only on data from the target robot domain. In Table II we show that the performance of Octo increases as we increase the number of training datasets. This suggests that expanding the data mix to even more datasets may further improve policy performance. We will leave this for future work, along with a more thorough investigation of best practices for data curation.

Training objective: We compare Octo’s diffusion decoding training objective (see Section III-C) to common alternatives from prior work: simple MSE loss [8, 47] and cross-entropy loss on discretized actions [10, 103]. In Table II we find that Octo’s diffusion training objective leads to substantially improved performance. This improvement is likely because the diffusion head can model multi-modal action distributions (unlike the MSE head) while retaining the precision of continuous actions (unlike the discrete head). Qualitatively, the policy acts more decisively than MSE-trained policies, and more precisely than those trained with discretized actions.

Model scale: We compare Octo models of three different

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Generalization honesty · Table VII WidowX

In-distribution

85%

Novel objects

80%

Novel environment

40%

Novel skill

5%

flip / insert
ANTI-HYPE

Cite 5% novel skill whenever someone says “generalist zero-shot”

Objects transfer; environments degrade; skills near floor — prevents foundation-model overclaim

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Octo = open OXE ViT+diffusion GRP bridge
Infrastructure history + eval baseline, not 2026 weights

Q&A · keep the spine clear

Octo: An Open-Source Generalist Robot Policy

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<https://octo-models.github.io>

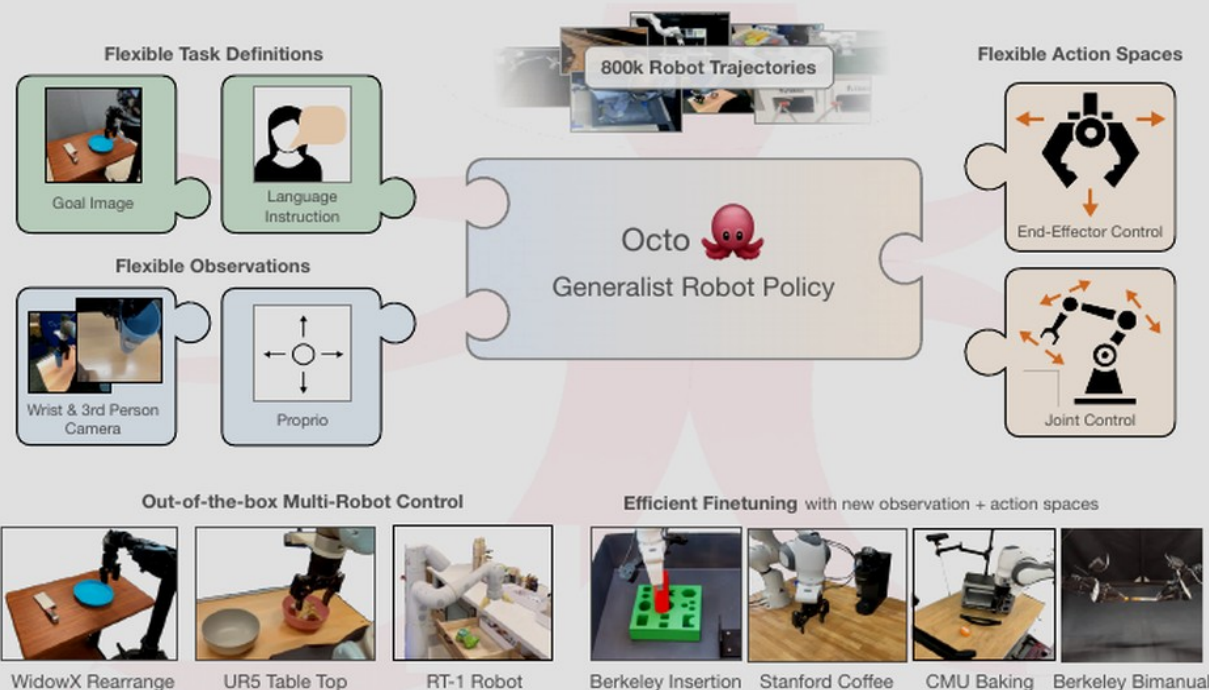


Fig. 1: We introduce Octo, an open-source, generalist policy for robotic manipulation. Octo is a transformer-based policy pretrained on 800k diverse robot episodes from the Open X-Embodiment dataset [67]. It supports flexible task and observation definitions and can be quickly finetuned to new observation and action spaces.

Abstract—Large policies pretrained on diverse robot datasets have the potential to transform robotic learning: instead of experiments across 9 robotic platforms, we demonstrate that Octo serves as a versatile policy initialization that can be effectively